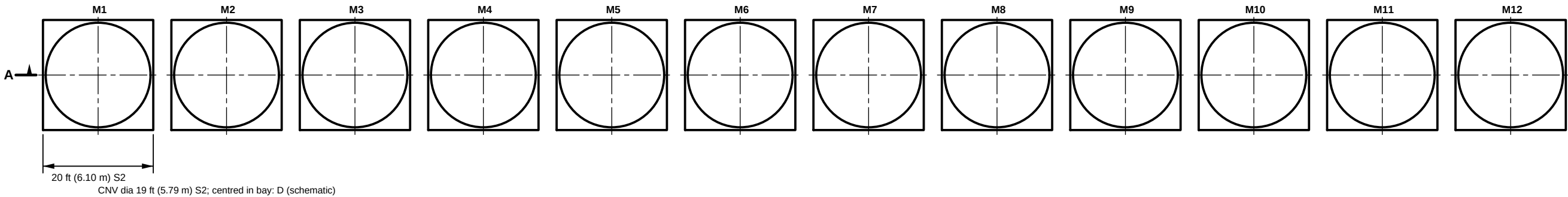
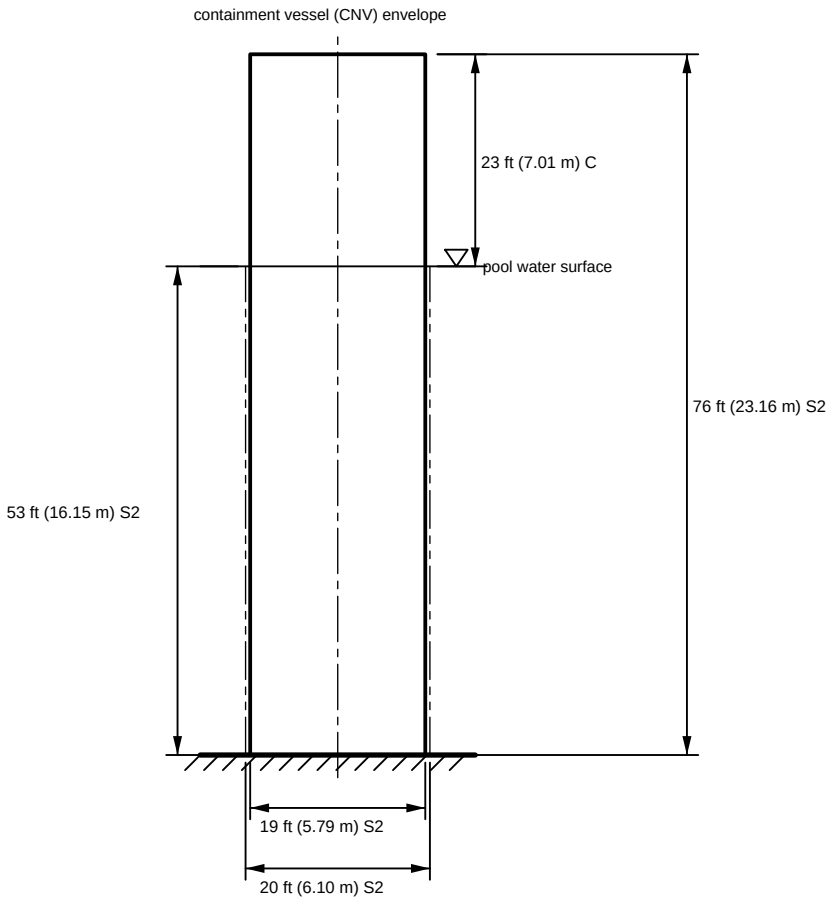


PLAN AT POOL LEVEL - MODULE BAYS (schematic order)

Each bay and containment at true size and scale. Bay ARRANGEMENT, spacing, walls, building outline and pool extent: UNKNOWN (G-dimensions) - bays shown in module order with a fixed sheet gap, NOT to scale between bays.



SECTION A-A - ONE MODULE BAY (true scale)



Radial clearance CNV to bay boundary: 0.5 ft (0.15 m) C (nominal; flanges, supports, guides not modelled)

CNV seated on the bay floor and centred: D (schematic placement; the support arrangement is not in the sources read).

NOT DRAWN - UNKNOWN (positions not in the sources read)

- RXB: Reactor building (Seismic Category I, below-grade pool)
- GALLERY: Reactor building steam gallery area (behind the module bays' shield walls)
- POOL: Common reactor pool / ultimate heat sink
- CR: Single control room for up to 12 modules
- ELEC: Electrical power (non-Class 1E AC and DC; no safety-related function relies on it)
- DHRS condensers (2 trains per module, submerged in the pool): positions UNKNOWN
- ECCS valves (3 RVV + 2 RRV per module, on the vessel): positions UNKNOWN
- shield walls between the bays and the steam gallery: positions UNKNOWN
- grade line: elevation UNKNOWN; rule: the pool portion of the building is below grade
- site layout outside the reactor building: not in the sources read

NOTES

SCREENING DRAWING of a public-source skeleton: not a plant drawing; not for construction, licensing or safety use.

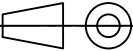
S2 = secondary, design-family value (Welter et al. 2023 also covers US460 / VOYGR): NOT a confirmed US600 dimension.

C = computed from drawn values; D = schematic placement; UNK = not drawn. Units ft (m in brackets).

Revision = hash of the facility model; any model change gives a new revision and supersedes this sheet.

Unknowns stay unknown until a human reads the FSAR sections on sheet 3.

PLANT GA - NUSCALE-US600 (public-source skeleton)

Drawing no. CK-GA-US600-PLANT Sheet 1 of 3	Revision
Scale 1:250 (A3) Units ft (m) Plan + section	R-9A68158C 
Model: nuscale-us600: 161 SSCs, 301 dependencies, 24 facts	model hash 9a68158c7c4fecf5e40e7ffb9213038c
SCREENING - SCHEMATIC - NOT FOR CONSTRUCTION	1d3ddedbc7abf5476ebd4f4889bafef2
Generated by ConKay facility-ga 1.0.0 from the facility model; revision = model hash	

Screening analysis only. Not a safety determination. Decisions and approvals belong to qualified engineers and regulators.

SSC REGISTER (from the model; per-module items grouped)

SSC / group	Scope	Count	Ids	Facts (verbatim quotes, sheet 3 / model)
Reactor building (Seismic Category I, below-grade pool)	shared	1	RXB	F-rxb, F-rxb-12, F-pool-grade
Reactor building steam gallery area (behind the module bays' shield walls)	shared	1	GALLERY	F-bay-gallery
Common reactor pool / ultimate heat sink	shared	1	POOL	F-pool, F-pool-heat
Single control room for up to 12 modules	shared	1	CR	F-cr
Electrical power (non-Class 1E AC and DC; no safety-related function relies on it)	shared	1	ELEC	F-no-1E, F-no-power
module n bay (in the pool, shield wall to the steam gallery)	per-module	12	M1..M12:BAY	F-bay-gallery, F-bay-size
power module n: RPV with core, pressurizer, two helical-coil SGs	per-module	12	M1..M12:NPM	F-module, F-rating
module n containment vessel	per-module	12	M1..M12:CNV	F-module, F-eccs
module n protection system	per-module	12	M1..M12:MPS	F-mps
module n DHRS train 1 isolation valves	per-module	12	M1..M12:DHRS1-VALVES	F-dhrs-act, F-failsafe
module n DHRS train 1 condenser (in the pool)	per-module	12	M1..M12:DHRS1-COND	F-dhrs, F-dhrs-pool
module n DHRS train 2 isolation valves	per-module	12	M1..M12:DHRS2-VALVES	F-dhrs-act, F-failsafe
module n DHRS train 2 condenser (in the pool)	per-module	12	M1..M12:DHRS2-COND	F-dhrs, F-dhrs-pool
module n ECCS reactor vent valve 1 (main, IAB, trip and reset valves)	per-module	12	M1..M12:RVV1	F-eccs, F-eccs-valve
module n ECCS reactor vent valve 2 (main, IAB, trip and reset valves)	per-module	12	M1..M12:RVV2	F-eccs, F-eccs-valve
module n ECCS reactor vent valve 3 (main, IAB, trip and reset valves)	per-module	12	M1..M12:RVV3	F-eccs, F-eccs-valve
module n ECCS reactor recirculation valve 1 (main, IAB, trip and reset valves)	per-module	12	M1..M12:RRV1	F-eccs, F-eccs-valve
module n ECCS reactor recirculation valve 2 (main, IAB, trip and reset valves)	per-module	12	M1..M12:RRV2	F-eccs, F-eccs-valve

DIMENSIONS (value, basis, source fact)

Module bay, square side (water)	20 ft (6.10 m) S2	F-bay-size [frontiers-2023, 1.1 Reactor configuration]: "Each NPM is operated in its own bay, immersed in a pool of water which is approximately 20 ft square by 53 ft deep (see Figure 3)."
Module bay water depth	53 ft (16.15 m) S2	F-bay-size [frontiers-2023, 1.1 Reactor configuration]: "Each NPM is operated in its own bay, immersed in a pool of water which is approximately 20 ft square by 53 ft deep (see Figure 3)."
Containment vessel height (nominal)	76 ft (23.16 m) S2	F-cnv-size [frontiers-2023, 1.1 Reactor configuration]: "The reactor vessel is enclosed in a stainless steel containment vessel (CNV) that is nominally 76 ft tall and 19 ft in diameter."
Containment vessel diameter (nominal)	19 ft (5.79 m) S2	F-cnv-size [frontiers-2023, 1.1 Reactor configuration]: "The reactor vessel is enclosed in a stainless steel containment vessel (CNV) that is nominally 76 ft tall and 19 ft in diameter."
CNV top above the pool surface	23 ft (7.01 m) C	computed: CNV height - bay water depth, with the CNV seated on the bay floor (D, schematic); consistent with the rule's "partially submerged" (F-pool), not a sourced elevation
Radial clearance, CNV to bay side	0.5 ft (0.15 m) C	computed: (bay side - CNV diameter) / 2 from the two S2 nominal values; flanges, supports and guides not modelled
Reactor building outline	UNKNOWN (G-dimensions)	not in the sources read
Bay arrangement / spacing	UNKNOWN (G-dimensions)	not in the sources read
Pool surface to grade	UNKNOWN (G-dimensions)	not in the sources read; stated only: F-pool-grade [fr-2023-00729, I. Background]: "The pool portion of the reactor building is located below grade."
Control room location	UNKNOWN (G-layout)	not in the sources read

CONFLICTS KEPT FOR REVIEW (not merged)

C-rvv-count: number of ECCS reactor vent valves per module: 3 (F-eccs, fr-2023-00729 (US600 rule, primary)) vs 2 (F-eccs-2rvv, frontiers-2023 (secondary; describes the later design family)); pending_human_review

NOTES

SCREENING DRAWING of a public-source skeleton: not a plant drawing; not for construction, licensing or safety use.

S2 = secondary, design-family value (Welter et al. 2023 also covers US460 / VOYGR); NOT a confirmed US600 dimension.

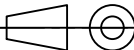
C = computed from drawn values; D = schematic placement; UNK = not drawn. Units ft (m in brackets).

Revision = hash of the facility model; any model change gives a new revision and supersedes this sheet.

Unknowns stay unknown until a human reads the FSAR sections on sheet 3.

Screening analysis only. Not a safety determination. Decisions and approvals belong to qualified engineers and regulators.

PLANT GA - NUSCALE-US600 (public-source skeleton)

Drawing no. CK-GA-US600-PLANT Sheet 2 of 3	Revision	
Scale 1:250 (A3) Units ft (m) Plan + section	R-9A68158C	
Model: nuscale-us600: 161 SSCs, 301 dependencies, 24 facts	model hash 9a68158c7c4fecf5e40e7ffb9213038c	
SCREENING - SCHEMATIC - NOT FOR CONSTRUCTION Generated by ConKay facility-ga 1.0.0 from the facility model; revision = model hash	1d3ddedbc7abf5476ebd4f4889bafef2	

FSAR GAPS - PULL LIST FOR A HUMAN (automated retrieval blocked; every gap stays UNKNOWN until read)

Chapter titles verbatim from 10 CFR 52 App. G III.A.2.b; accessions from 88 FR 3287 XVII. Section basis: rule = cited in the rule; verify = standard-format number, check the DCA contents.

Gap	Document	ADAMS	Chapter (Tier 2)	Section / figure	Basis
G-eccs-success	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Six, Engineered Safety Features	6.3 emergency core cooling system	verify
	NuScale DCA Final Safety Evaluation Report, August 2020	ML20023A318	Chapter Fifteen, Transient and Accident Analyses	15.0.0.5, "Limiting Single Failures"	rule
	NuScale DCA Final Safety Evaluation Report, August 2020	ML20023A318	Chapter Nineteen, Probabilistic Risk Assessment and Severe Accident Evaluation	19.1.4.6.4, "Success Criteria, Accident Sequences, and Systems Analyses"	rule
	SRM-SECY-19-0036, Staff Requirements--SECY-19-0036--Application of the Single Failure Criterion to NuSca...	ML19183A408	-	whole document	rule
G-dhrs-valves	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Five, Reactor Coolant System and Connecting Systems	the decay heat removal system section	verify
	NuScale DCA Final Safety Evaluation Report, August 2020	ML20023A318	Chapter Five, Reactor Coolant System and Connecting Systems	the decay heat removal system section	verify
G-dc	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Eight, Electric Power	8.3.2 DC power systems	verify
	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Twenty-One, Multi-Module Design Considerations	shared electrical systems	verify
	NRC Safety Evaluation for NuScale Power, LLC, Topical Report, TR-0815-16497, Revision 1, Safety Classifi...	ML17340A524	-	whole document	rule
G-mps	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Seven, Instrumentation and Controls	module protection system	verify
G-layout	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter One, Introduction and General Description of the Plant	1.2 general plant description (arrangement figures)	verify
	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Twelve, Radiation Protection	Figure 12.3-1, "Reactor Building Radiation Zone Map"	rule
	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Nine, Auxiliary Systems	9.5.1 fire protection (fire areas)	verify
	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Twenty-One, Multi-Module Design Considerations	shared structures and spatial interactions	verify
G-pool-inventory	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Nine, Auxiliary Systems	9.2.5 ultimate heat sink	verify
	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Twenty-One, Multi-Module Design Considerations	shared pool	verify
G-dimensions	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter One, Introduction and General Description of the Plant	1.2 general arrangement drawings of the reactor building	verify
	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Three, Design of Structures, Systems, Components and Equipment	3.8 seismic Category I structures (reactor building, pool)	verify
	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Twelve, Radiation Protection	Figure 12.3-1, "Reactor Building Radiation Zone Map"	rule
G-reliability	NuScale Power, LLC, Submittal of the NuScale Standard Plant Design Certification Application, Revision 5...	ML20225A071	Chapter Nineteen, Probabilistic Risk Assessment and Severe Accident Evaluation	19.1 probabilistic risk assessment (data)	verify
	NuScale DCA Final Safety Evaluation Report, August 2020	ML20023A318	Chapter Nineteen, Probabilistic Risk Assessment and Severe Accident Evaluation	19.1	verify

GAP EFFECTS ON THIS DRAWING AND THE SCREEN

G-eccs-success: ECCS success criterion (how many RRVs and RRVs must open) -> ECCS is screened only for supports every valve train shares (valid for any success criterion); no k-of-n single-failure screen

G-dhrs-valves: DHRS actuation / isolation valve count and type per train -> each train is modelled as one valve group + one condenser; no NUREG/CR-6928 data row can be matched

G-dc: non-Class 1E DC system architecture (common vs module-specific, divisions, battery duration) -> multi-module actuation on loss of a shared DC supply is not determinable

G-mps: module protection system separation groups / divisions and their power -> MPS modelled as one actuating system per module; channel independence not screened

G-layout: building, room and fire-area layout of DHRS / ECCS equipment -> spatial common-cause (fire, flood) not screened

G-pool-inventory: pool inventory, level limits and heat-up / boil-off times for the US600 -> the shared heat sink is a structural dependency only; no time to boil

G-dimensions: US600 building, pool, bay and containment dimensions and the bay arrangement -> the facility GA is schematic: the only dimensions drawn are the secondary, design-family values (CNV 76 ft x 19 ft, bay 20 ft square x 53 ft deep), labelled S2; the bay arrangement, building outline and pool-to-grade elevation are UNKNOWN

G-reliability: failure data for passive components (condensers, pool, building) and NuScale valve designs -> fault and event trees are built and their cut sets computed; nothing is quantified

RETRIEVAL LOG (automated attempts)

403 <https://www.nrc.gov/reactors/new-reactors/smr/licensing-activities/nuscale.html>

403 <https://www.nrc.gov/docs/ML2022/ML20225A071.html>

403 <https://pbadupws.nrc.gov/docs/ML2022/ML20225A071.html>

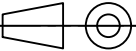
no response (20 s timeout) <https://adams.nrc.gov/wba/services/search/advanced/nrc> (ML20225A071 query)

401 <https://adams-api.nrc.gov/aps/api/search/ML20225A071> (the ADAMS API needs a registered subscription key)

200 <https://www.govinfo.gov/content/pkg/FR-2023-01-19/html/2023-00729.htm> (the final rule: read; quotes and locators re-checked against it)

200 <https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2023.1160150/full> (the secondary review: read)

PLANT GA - NUSCALE-US600 (public-source skeleton)

Drawing no. CK-GA-US600-PLANT Sheet 3 of 3	Revision
Scale 1:250 (A3) Units ft (m) Plan + section	R-9A68158C 
Model: nuscale-us600: 161 SSCs, 301 dependencies, 24 facts	model hash 9a68158c7c4fecf5e40e7ffb9213038c
SCREENING - SCHEMATIC - NOT FOR CONSTRUCTION Generated by ConKay facility-ga 1.0.0 from the facility model; revision = model hash	1d3ddedbc7abf5476ebd4f4889bafef2